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Dynamic Path Optimization of Cold Chain Logistics Based on Improved Ant Colony Algorithm from a Low-Carbon Perspective

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Abstract: Cold chain logistics face challenges such as high energy consumption, high carbon emissions, and high rates of cargo spoilage during transportation. These issues increase the operating costs of enterprises and also negatively affect the ecological environment. Therefore, this study constructs a cold chain logistics distribution path optimization model that comprehensively considers time window constraints, carbon emission costs, and cargo damage costs. Based on this, an I-ACO algorithm is proposed to achieve low-carbon and efficient distribution path planning. The I-ACO algorithm enhances the accuracy and adaptability of path selection by introducing dynamic time warping technology and chaos theory, thereby improving the search ability and solution quality of the algorithm. The results indicated that the I-ACO had a very high overall sample fitting degree, with a correlation coefficient of 0.99 between the actual output value and the expected value. Its accuracy reached 95.2%, with fast convergence speed and a carbon emission of only 0.04 kg, significantly lower than other algorithms. The I-ACO could effectively reduce logistics costs, decrease carbon emissions, and improve delivery efficiency and customer satisfaction. The research algorithm provides an efficient, accurate, and environmentally friendly solution for optimizing the distribution path of cold chain logistics. This is beneficial for cold chain logistics enterprises to achieve economic benefits and better fulfill social responsibilities, and promote the sustainable development of the industry.

Keywords: Cold chain logistics; Dynamic path planning; Ant colony optimization algorithm; Carbon emissions

1. Introduction

With the growth of the global economy and the rising demand for fresh food, the Cold Chain Logistics (CCL) industry has also ushered in opportunities for rapid development. As a key link in ensuring the safety of perishable goods, including food and medicine, the CCL importance is increasingly prominent [1-2]. In recent years, the rise of fresh food e-commerce and people's pursuit of a high-quality life have led

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to a continuous increase in demand for CCL. According to relevant data, the global CCL market has maintained a double-digit growth rate in the past decade and is hoped to maintain strong growth momentum in the coming years. CCL involves the transportation of food and medicine, and covers multiple fields such as biological products, flowers, electronic components, etc. Its application scenarios are wide, and the market demand is huge [3-4]. However, the traditional CCL delivery model often overlooks the impact of Carbon Emissions (CEs) on the environment, resulting in high energy consumption and pollution during transportation. During the transportation of CCL, a large amount of refrigeration equipment is required, which consumes a significant amount of energy and generates a lot of CEs, causing serious environmental impacts. In addition, there are also issues with high rates of goods spoilage and low customer satisfaction in the distribution process of CCL, which increases the operating costs of the enterprise and seriously affects its economic and social benefits [5-6].

With the proposal of the "dual carbon" target, the low-carbon development of the CCL industry has become an inevitable trend. Zhao put forth an improved fireworks Artificial Bee Colony Optimization (ABCO) algorithm to optimize the delivery path in response to environmental pollution and high expenditures in CCL distribution. The benchmark function verification showed that this algorithm was superior in solving the optima, mean, and standard deviation. In a simulation analysis of a fresh food firm, all the delivery expenditures were lowered by a mean of 3.30% when CEs were considered, and by 4.93% when not considered. This algorithm was more benefical for cutting down the quality loss, pollutant emissions, and CEs [7]. Benyamin et al. designed a mixed-integer nonlinear programming model based on the balanced scorecard for the optimization problem of the CCL system. This model considered

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customer satisfaction, unit cost, and greenhouse gas emissions to optimize the logistics and distribution system for marine products. This model could accurately evaluate the cost of CCL and help managers choose the optimal number of distribution centers and efficient routes, reduce system costs, and balance environmental and economic benefits [8]. Abdullah et al. proposed a rough Back Propagation-Particle Swarm Optimization (BP-PSO) network model to address the issue of selecting the best CCL service provider for cold chain food production enterprises. Through numerical verification, the prediction error of this model was 40.94% lower than that of BP, and the prediction results were more precise and reliable [9]. Zhang et al. proposed a hybrid Ant Colony Optimization (ACO) based on the elbow rule to address balancing economic costs and environmental benefits in CCL. This algorithm outperformed ordinary ACO in convergence velocity and solution quality, providing an important reference for the CCL company to balance costs and benefits and solve the problem of CCL's distribution of agricultural goods [10].

With the rapid development of e-commerce, logistics companies are facing challenges in efficient route planning and scheduling. To meet the dynamic distribution demands, Guo J et al. proposed a new logistics scheduling model, utilized radio frequency identification technology to optimize vehicle routes, and developed a vehicle scheduling model. By combining the adaptive tabu search algorithm and the nearest neighbor algorithm, the problems of random customer demand and service time have been solved. Comparative experiments showed that the improved method outperformed other methods in task completion and queue time reduction [11]. Aiming at the issue of optimizing CCL distribution routes, Shang W et al. proposed a low-carbon distribution route optimization model based on Internet of Things

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technology. By using genetic algorithms as the path optimization tool, the optimal or nearly optimal distribution path was approximated through iterative evolution, effectively reducing CEs. The experimental results showed that the optimized distribution route reduced the total driving distance by an average of 10.2%, the total energy consumption by 12.5%, and the total CEs by 12.8%, while also reducing the temperature fluctuation range by an average of 13.3% [12]. To address the problems of low efficiency and high operating costs in logistics route optimization methods, Almazroi et al. proposed the Hybrid Firefly-Spotted Hyena Optimization (HF-SHO) algorithm. The research results showed that the HF-SHO algorithm performed excellently in logistics path optimization, with the minimum path distance reaching 546 units, emphasizing the importance of choosing the appropriate algorithm in logistics path navigation [13].

In summary, although significant progress has been made in CCL path optimization, there are still some shortcomings. Existing research mostly focuses on single objective optimization, such as cost or CEs, while there is relatively less research on multi-objective comprehensive optimization. Secondly, some studies lack in-depth analysis of practical application scenarios, resulting in limited applicability of models and algorithms. In addition, there is insufficient consideration for the dynamic changes and real-time updates of customer needs, especially in the more complex scenarios of multiple distribution centers and multiple vehicle models, where adaptability still needs further verification. Given these issues, this study establishes a CCL distribution path optimization model based on a Low-Carbon Perspective (LCP) and designs an Improved ACO (I-ACO) solution method. This model comprehensively considers Time Window Constraints (TWCs), CE Costs (CECs), and Cargo Damage Costs (CDCs), aiming to achieve a low-carbon, efficient, and

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economic balance in CCL distribution. The I-ACO algorithm enhances the accuracy and adaptability of path selection by introducing Dynamic Time Warping (DTW) technology and chaos theory, optimizes the quality of the initial population, thereby improving the search ability and solution quality. The innovation of the research lies in combining low-carbon concepts with CCL path optimization, proposing a comprehensive multi-objective optimization model. Moreover, through the I-ACO algorithm, an effective response to complex dynamic environments has been achieved. The contribution of the research lies in the in-depth study of the optimization problem of CCL distribution routes by combining DTW technology with chaos theory. By introducing DTW technology, the algorithm can more accurately evaluate the efficiency and cost of different distribution routes, thereby selecting the optimal one. Meanwhile, by leveraging chaos theory to generate an ordered initial population structure, the search ability and solution quality of the algorithm have been effectively enhanced. This comprehensive approach offers a more targeted and efficient solution for optimizing CCL distribution routes, significantly enhancing the accuracy and adaptability of route selection, and providing new ideas and methods for the sustainable development of the CCL industry.

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2. Methods and Materials

In the context of "dual carbon", this study analyzes the influencing factors of CEs in the delivery process of CCL, and constructs a CCL delivery path optimization model that comprehensively considers time windows and CEs to address the problems existing in CCL delivery. At the same time, a solution method based on the I-ACO algorithm is designed, which optimizes the delivery path, reduces the total logistics cost, and reduces pollutant emissions by introducing DTW technology and chaos theory improvements.

2.1 Construction of CCL delivery path optimization model under LCP

Against the backdrop of the "dual carbon" goal, the CCL distribution industry is facing new challenges and opportunities. Currently, there are multiple issues in the delivery process of this industry, including low on-time delivery rates, high product spoilage rates, and severe environmental pollution. These issues compress the profit margins of enterprises and have adverse effects on the ecological environment. Among them, excessive energy consumption in the logistics process has caused serious ecological damage, which has led government departments to increase CE control over enterprises. At the same time, unreasonable delivery arrangements prevent goods from arriving at their destination on time, leading to a reduction in customer resources and further exacerbating the operational pressure on the enterprise [14-15]. Table 1 shows the temperature range required for different cold chain goods.

Table 1 Temperature ranges required for different CCL goods

Categories	Product type	Temperature requirement
Fresh-keeping articles	Vegetables, flowers	2 to 8 ℃
	Fruit	
	Fresh aquatic products	
	Electronic component	
Chilled goods	Acid meat	0 to 5 ℃
	Soybean products	
	Chocolate	
Frozen goods	Quick-frozen food	-10 to -18 ℃

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	Frozen meat products	
	Frozen fresh seafood products	
Deep-frozen items	Premium ice cream	-20 to -45 ℃
	Highly dangerous goods	
	Premium bread live yeast dough	

The temperature range provided in Table 1 is to ensure that various products maintain their freshness and quality during transportation and storage. In the logistics and distribution process, CEs are mainly composed of two factors: CO_2 released by operating the refrigeration system in the Cold Chain Storage (CCS) process; CEs from vehicle exhaust during transportation and the energy consumption of refrigeration equipment during transportation. Reducing greenhouse gas emissions, promoting the sustainable development of CCL, and ensuring that product quality is not affected or further optimized have become the core issues that the industry urgently needs to address. Figure 1 shows the operation flowchart of CCL.

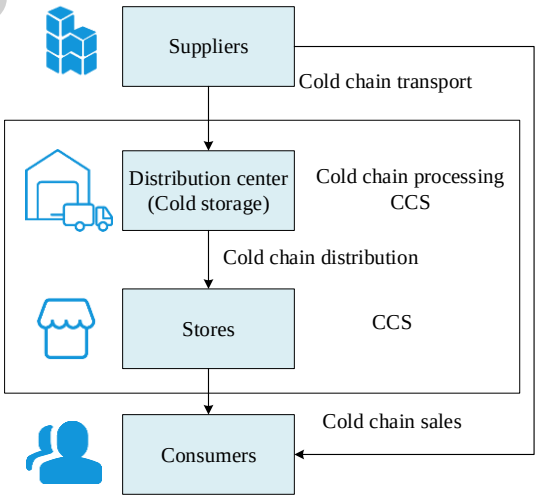

Figure 1 CCL operation flow chart

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In Figure 1, the goods depart from the supplier and are transported via cold chain to the distribution center (CCS), where they undergo cold chain processing and storage. Subsequently, the goods are delivered to various stores through the CCL distribution process, and finally reach consumers through the cold chain sales process. The entire process emphasizes the importance of low-temperature control throughout the entire process to ensure the freshness of goods during transportation and storage. Figure 2 shows the vehicle routing problem.

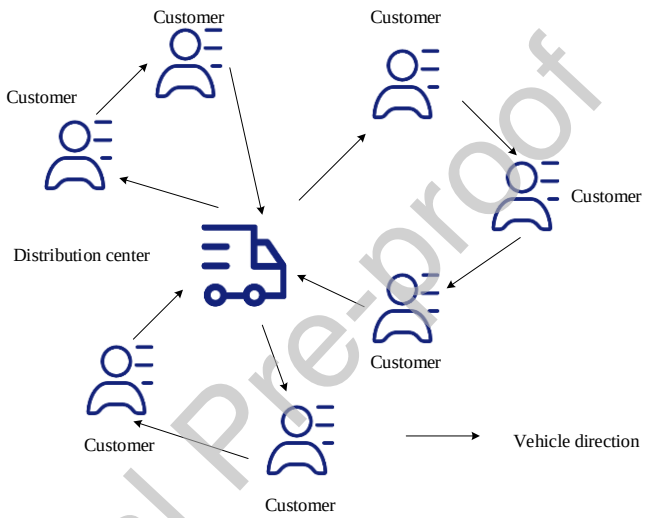

Figure 2 Diagram of the vehicle path

In Figure 2, arrows indicate the specific direction of travel for vehicles departing from the distribution center to reach various customers. Optimizing vehicle travel routes to reduce transportation costs can enhance delivery efficiency and perfect customer satisfaction. Planning the route reasonably can ensure that vehicles can efficiently access each customer, thereby optimizing CCL delivery. Finally, this study constructs a CCL vehicle path planning model that comprehensively considers time windows and CEs. This model needs to consider customers' requirements for the delivery time of goods and includes fuel consumption and CEs from refrigeration equipment as CECs in the overall objective function. The objective function integrates transportation costs, CECs, and CDCs, and the specific mathematical model is given

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by equation (1).

$$\begin{cases} \min Z = \alpha C_{trans} + \beta C_{emission} + \gamma C_{damage} \\ \text{s.t.} \quad \sum_{i=1}^{n} x_{ij} = 1, w_j \cdot x_{ij} \le Q, a_j \le t_j \le b_j \end{cases} \quad (1)$$

In equation (1), $\min Z$ is the minimum of the objective function. C_{trans} is the total transportation cost. $C_{emission}$ is the total CECs. C_{damage} is the total cost of goods damage. α, β, and γ are weights. w_j represents the weight, and Q is the maximum load capacity of the vehicle. a_j and b_j represent the lower and upper limits of the time window for customers to point j. t_j represents the time when the vehicle arrives at the customer's point j. Firstly, based on the enterprise's emphasis on cost control, a relatively high weight is assigned to transportation costs α. Secondly, in light of the enterprise's carbon reduction targets and environmental protection policy requirements, appropriate weights should be given to the CECs β. Finally, the cost of goods damage is determined based on the customer's requirements for the quality and integrity of the goods γ. The allocation of weight parameters needs to be completed through internal decision-making meetings, consulting with industry experts, or analyzing historical data to ensure that the model can simultaneously meet the economic benefits of the enterprise and balance environmental and social benefits. The calculation of the total transportation cost is shown in equation (2).

$$C_{\text{trans}} = \sum_{i=1}^{n} \sum_{j=1}^{n} d_{ij} \cdot x_{ij} \quad (2)$$

In equation (2), d_{ij} represents the distance between node i and node j. x_{ij} is a decision variable. Its value is 1 when the vehicle travels from node i to node j, and 0 otherwise. The calculation of the total CEC is shown in equation (3).

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$$C_{\text{emission}} = \sum_{i=1}^{n}\sum_{j=1}^{n} e_{ij} \cdot x_{ij} \quad (3)$$

In equation (3), e_{ij} represents the CEs of the vehicle from node i to node j, which are related to the vehicle's fuel consumption rate and driving distance. The calculation of the total cost of cargo damage is shown in equation (4).

$$C_{\text{damage}} = \sum_{i=1}^{n}\sum_{j=1}^{n} c_{ij} \cdot x_{ij} \quad (4)$$

In equation (4), c_{ij} represents the possible loss cost of goods during transportation from node i to node j, which is related to the transportation time and the characteristics of the goods. The constraints of the model include: each customer point must be visited exactly once; The total weight of each delivery route must not exceed the maximum load capacity of the vehicle; The delivery time must meet the customer's time window requirements. These constraints ensure the feasibility and effectiveness of the model in practical applications.

2.2 Optimization of CCL delivery path based on I-ACO algorithm

On the basis of constructing the CCL delivery path optimization model under LCP, a path optimization method based on I-ACO is further constructed. The I-ACO algorithm enhances the accuracy and adaptability of path selection by introducing DTW technology and chaos theory, and optimizes the quality of the initial population. This provides a more targeted and efficient solution for optimizing CCL delivery routes [16-17]. This algorithm not only considers the importance of pheromones and the visibility of inter-node paths but also introduces DTW to enhance the accuracy and adaptability of path selection. The probability of each ant transitioning from nodes i to j after t iterations is given by equation (5).

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$$p_{ij}^{k}(t)=\begin{cases}\dfrac{\tau_{ij}^{\alpha}\cdot\eta_{ij}^{\beta}}{\sum_{j\in allowed_k}\tau_{ij}^{\alpha}\cdot\eta_{ij}^{\beta}}, j\in allowed_k \\ 0, other\end{cases} \quad (5)$$

In equation (5), α and β are weight factors for pheromone and path visibility. τ_{ij} means the concentration of pheromones between i and j. η_{ij} is a heuristic factor between i and j. $allowed_k$ denotes the set of nodes accessible to the k-th ant in the current iteration. Heuristic information η_{ij} is defined as the reciprocal of the distance between nodes, as shown in equation (6).

$$\eta_{ij}=\frac{1}{d_{ij}} \quad (6)$$

In equation (6), d_{ij} is the Euclidean distance among nodes. τ_{ij} will update as the iterations increase, as shown in equation (7).

$$\begin{cases}\tau_{ij}(t+1)=(1-\rho)\tau_{ij}+\Delta\tau_{ij}(t) \\ \Delta\tau_{ij}(t)=\sum_{k=1}^{m}\Delta\tau_{ij}^{k}(t)\end{cases} \quad (7)$$

In equation (7), ρ is the volatility coefficient of pheromones. $\Delta\tau_{ij}^{k}(t)$ is the pheromone increment left by the k-th ant between i and j after t iterations. To further optimize path selection, this study introduces DTW technology to calculate the similarity between 2 time series. DTW measures their similarity by adjusting the alignment of time series to minimize the distance between them [18-19]. The DTW distance between two time series is defined as shown in equation (8).

$$D(L_1,L_2)=DTW=\sqrt{d_{m,n}} \quad (8)$$

In equation (8), $D(L_1,L_2)$ is the DTW distance between two time series L_1 and L_2, used to evaluate the similarity between these two sequences. In this way, the algorithm can more accurately evaluate the efficiency and cost of different delivery

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routes, and thus select the optimal path. In addition, the algorithm also considers CEs and TWCs to achieve low-carbon and efficient CCL distribution. The calculation steps for the DTW distance between two time series L_1 and L_2 are shown in Figure 3.

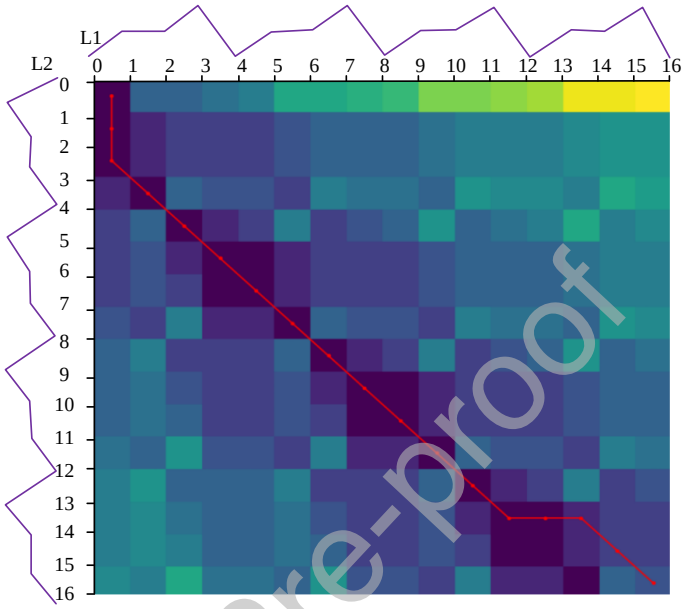

Figure 3 Diagram of the solution process of DTW distance

In Figure 3, the two lines represent the trends of data changes for two time series. In the matrix below, the horizontal and vertical axes correspond to various time points of the two time series. The color depth of each cell in the matrix represents the square of the difference between two sequence values at the corresponding time point. The darker the color, the greater the difference. The red line in the figure represents the best matching path between two sequence time points when calculating the DTW distance. Through this path, the minimum distance between two sequences, known as the DTW distance, can be found to measure their similarity. When applying the basic ACO algorithm to solve optimization problems, the random generation of the initial population may potentially limit the heterogeneity of the population, thereby adversely affecting the quality of the final optimization results. To construct a starting

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group with better characteristics, chaos theory is adopted, and the characteristics of chaotic systems are utilized to generate an ordered starting group structure. Chaos theory is an ideal tool for solving optimization combinatorial problems, which can effectively screen out high-quality initial populations [20-21]. Among numerous chaotic models, logistic mapping is a highly representative one-dimensional chaotic system, expressed as equation (9).

$$X(t+1) = \mu X(t)(1 - X(t)) \quad (9)$$

In equation (9), μ is a parameter that affects the speed and direction of system state changes. $X(t)$ and $X(t+1)$ are the system state variable indicated at time t and $t+1$. The graph of the dynamic changes of the logistic mapping system under different parameter settings is shown in Figure 4.

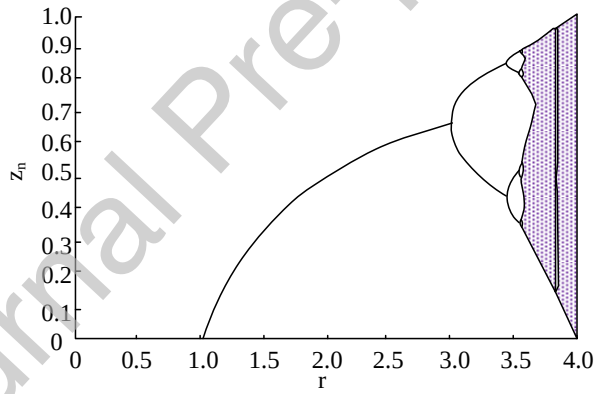

Figure 4 Logistic mapping system dynamic changes under different parameter settings

In Figure 4, when the parameters are small, the system state variables gradually stabilize over time, showing a smooth growth curve. However, when the parameters exceed a certain critical value, the system state variables begin to exhibit periodic fluctuations, and the amplitude of the fluctuations gradually increases, eventually entering a chaotic state. The shaded area in the figure emphasizes the complex dynamic behavior of the system in chaotic states, indicating the sensitivity of the system to initial conditions [22-23]. The flowchart of the I-ACO algorithm is shown

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in Figure 5.

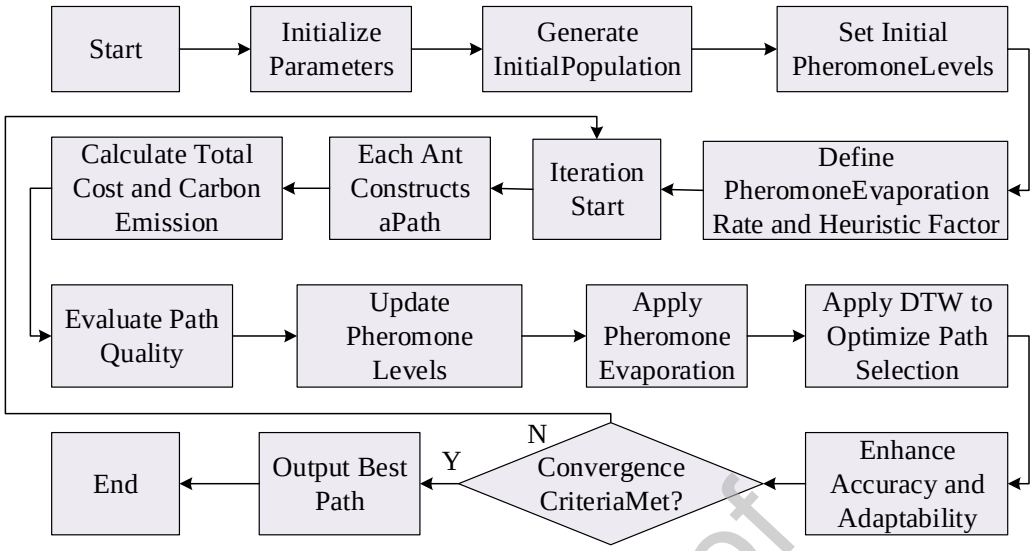

Figure 5 Flowchart of the I-ACO algorithm

In Figure 5, the algorithm parameters are initialized, including population size, pheromone volatility, and heuristic factors. Next, the initial solution set is randomly generated, and the pheromone concentration is set. During the iterative process, each ant builds a path based on a probability model and calculates the total cost and CEs of each path. Subsequently, the pheromone concentration is updated based on the path quality, and the pheromone evaporation rule is applied to simulate the natural dissipation of pheromones over time. In addition, DTW technology is introduced to optimize path selection and enhance the accuracy and adaptability of the algorithm. This process is repeated until the convergence condition is met, and finally, the optimal path is output as the solution. The pseudo-code for improving the ACO algorithm is shown as follows:

Algorithm: I-ACO

Input: Problem parameters

1. Initialize parameters:

    - Set the number of ants (number_of_ants)

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```
    - Set the pheromone evaporation rate (evaporation_rate)
    - Set the heuristic factor (heuristic_factor)
    - Set the maximum number of iterations (max_iterations)
    - Set the carbon emission coefficient (carbon_emission_coefficient)
    - Initialize the pheromone matrix (pheromone_matrix) with a uniform small value
2. Initialize the population:
    - Use chaos theory to generate initial solutions for the population
3. for iteration from 1 to max_iterations do
4.      for each ant ant in colony do
5.          Construct a route path using probability selection rules (considering pheromone and heuristic information)
6.          Calculate the total cost total_cost and carbon emission carbon_emission of the route
7.      end for
8.      Evaluate all routes:
9.          for each route path in all_routes do
10.             Use DTW technology to optimize route selection
11.             Update the pheromone matrix pheromone_matrix using the total cost and carbon emission of the route
12.         end for
13.     Apply pheromone evaporation evaporation_rate to the pheromone matrix
14.     if convergence criteria are met then
15.         break
16.     end if
17. end for
18. Return the best route
```

Finally, the optimization process of CCL delivery path based on I-ACO is shown in Figure 6.

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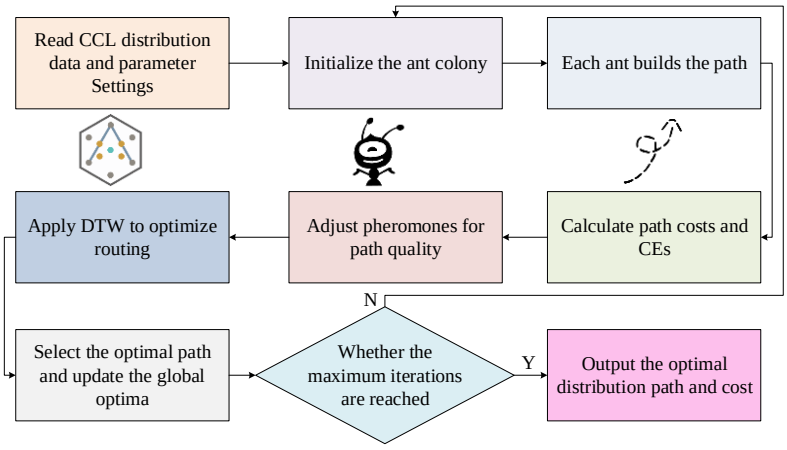

Figure 6 Flowchart of CCL distribution route optimization based on I-ACO algorithm

In Figure 6, the process first reads CCL delivery data and parameter settings, inputs distribution center location, customer location and demand, vehicle capacity and speed, TWCs, and CE factors. Step 2 is to initialize the ant colony, randomly generate an initial solution set, set the initial pheromone concentration, define the pheromone volatilization rate, and heuristic factors. After the iteration process begins, each ant selects the next node to construct a path based on probability. Step 4 is to calculate the overall expenditure of each route, containing transportation costs, CECs, and CDCs. Step 5 is to evaluate the CEs of the pathway, and then update the pheromone concentration based on the pathway quality, applying pheromone volatilization and update rules. The final step is to apply DTW technology to optimize path selection, enhance the accuracy and adaptability of path selection.

3. Results

This study validates the performance of the I-ACO in optimizing CCL delivery paths through experiments. Comparisons are made between the improved algorithm and other common optimization algorithms under different datasets and testing conditions. The performance of the algorithm on key indicators such as accuracy, convergence speed, CEs, total cost, path length, customer satisfaction, and vehicle

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utilization is evaluated to validate the effectiveness of the I-ACO in CCL delivery path optimization tasks.

3.1 Performance evaluation of I-ACO algorithm

To verify the validity of the proposed model, the study adopts the widely recognized public standard dataset, the Solomon dataset, which is frequently used in vehicle routing problem research. The selected dataset contains instances of distribution problems of different scales, specifically including various scenarios ranging from 10 to 100 customer points. These examples consider the basic distribution requirements and cover complex factors such as TWCs and vehicle capacity limitations. When conducting numerical experiments, 70% of the data is used as the training set and 30% as the test set to verify the generalization ability. This study validates the effectiveness of the I-ACO in optimizing low-carbon CCL delivery routes, as well as its performance in reducing CEs and overall costs. The experimental environment selects software Matrix Laboratory (MATLAB) 2017b, database My Structured Query Language (MySQL) 8.0, development framework Spring Boot 2.2.5, and persistence layer framework Mybatis. Spring Boot is used as the development framework to build back-end services, while MySQL 8.0 serves as the database management system for storing and processing the data required for experiments. Selecting these platforms aims to ensure the efficiency of algorithm implementation and the reliability of data processing. The constructed test framework includes a simulation environment that can evaluate the I-ACO algorithm and verify the algorithm's performance on different datasets through interaction with the MySQL database. In addition, this framework also supports quantitative analysis of key indicators such as the convergence speed, accuracy rate, and customer satisfaction of the algorithm, thereby providing comprehensive testing and verification for the

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effectiveness. Table 2 shows the specific experimental parameters.

Table 2 Experimental parameter configuration table

Parameter name	Parameters	Units
Population size	50	/
Pheromone volatile factor	0.6	/
Heuristic factor	5	/
Maximum iterations	500	Time (s)
Vehicle speed	60	km/h
Maximum vehicle mileage	120	km
Maximum load of vehicle	4000	kg
Lead penalty factor	150	Yuan per hour
Delay penalty factor	250	Yuan per hour
No-load fuel consumption	0.18	/
Full load fuel consumption	0.40	/
Product freshness attenuation	0.45	/
Discharge refrigerant consumption	12	Yuan per hour
Carbon tax	20	Yuan/ton
CE coefficient	3.10	kg-co_2/kg
CE coefficient of refrigeration	2.60	kg-co_2/kg

In Table 2, the population size of the ant colony is 50, the pheromone volatilization factor is 0.6, the heuristic factor is 5, and the maximum iterations are

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500. The vehicle speed is set to 60 km/h, the maximum mileage is 120 km, and the maximum load is 4,000 kg. The penalty coefficients for early arrival and delayed arrival are 150 yuan/h and 250 yuan/h, respectively. There are differences in fuel consumption rates under different load conditions: 0.18 at no load and 0.40 at full load. The coefficient of decline in product quality is 0.45, the cost of refrigerant usage during unloading is 12 yuan/h, and the CE tax is 20 yuan/t. The experiment uses a validated public standard dataset. These datasets are widely used in the research of CCL distribution route optimization problems, ensuring the rationality of numerical selection. In addition, the parameter settings used in the experiment are all based on the real-world CCL operation standards and industry best practices. The selection of these parameters aims to simulate real-world distribution scenarios, thereby making the evaluation results of the model closer to reality and enhancing the practical relevance and application value of the research. Firstly, the regression performance of the I-ACO on the total sample, training set, and test set is shown in Figure 7.

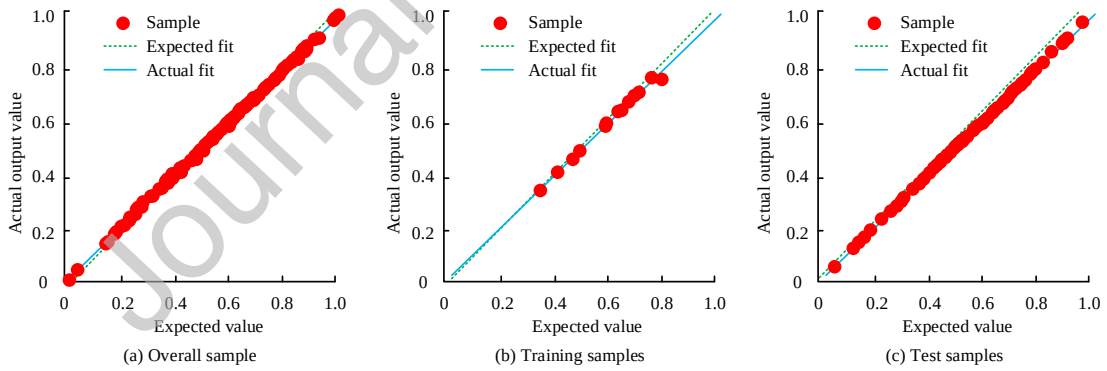

Figure 7 Regression of the I-ACO algorithm

In Figure 7 (a), the overall fitting degree of the sample is very high, and the Correlation Coefficient (CorrC) between the actual output and the expected value reaches 0.99. This indicates that the I-ACO algorithm can predict output values very accurately, with almost no errors in the model. In Figure 7 (b), the fitting degree on the training set is also very high, with a CorrC of 0.98. This indicates that the I-ACO

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can learn data features well during the training phase and make very accurate predictions on the training data. In Figure 7 (c), the fitting degree on the test set is similar to the previous two, with a CorrC of 0.97. The I-ACO performs well on the training set and keeps high prediction accuracy on unseen test data. Next, this study will compare the I-ACO with ACO, Hybrid Ant Colony Optimization (HACO) [10], and ABCO. Figure 8 shows the convergence of several algorithms.

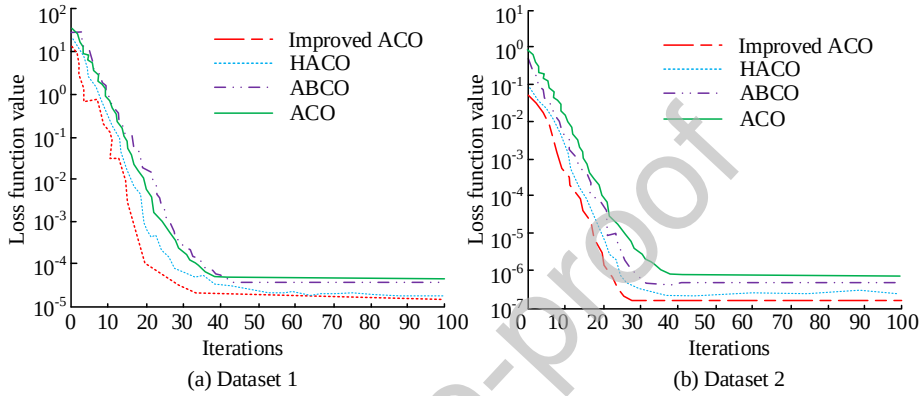

Figure 8 Convergence of several algorithms

In Figure 8 (a), on dataset 1, the I-ACO algorithm has already reduced the loss function value to below 10^{-5} after 20 iterations, and continues to exhibit the fastest convergence velocity throughout the entire iteration process. The convergence velocity of HACO and ABCO is second, while that of ACO is the slowest, and it is not until 60 iterations that the loss function value drops below 10^{-5}. In Figure 8 (b), on dataset 2, the I-ACO algorithm also exhibits the fastest convergence speed, with the loss function value dropping below 10^{-6} after about 30 iterations. The convergence speed of HACO and ABCO is similar, with ACO having the slowest convergence speed until its loss function value drops below 10^{-6} after 80 iterations. The I-ACO algorithm exhibits excellent convergence performance on both datasets, with fast convergence speed and lower final loss function values. This indicates that the I-ACO algorithm can effectively improve the search capability and resolution quality of the

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algorithm when dealing with CCL delivery path optimization problems, thereby finding better solutions faster. The accuracy and Receiver Operating Characteristic (ROC) curve of several algorithms are shown in Figure 9.

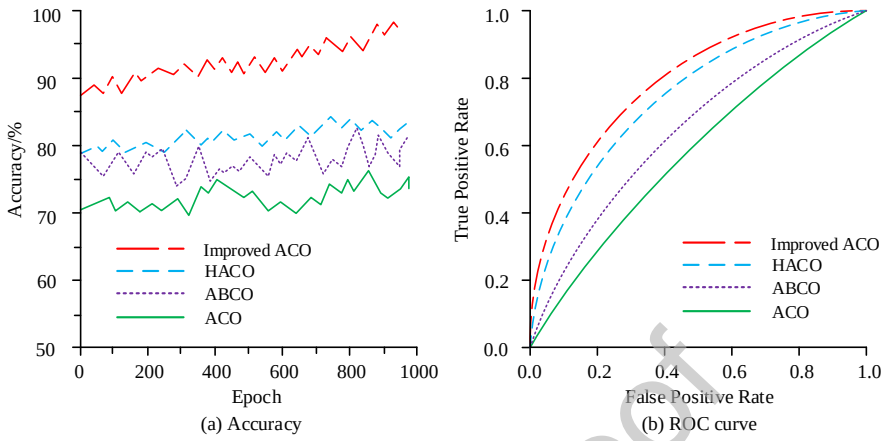

Figure 9 Accuracy and ROC curves of several algorithms

In Figure 9 (a), the I-ACO consistently has the highest accuracy, showing a rapid growth rate from the early stages of training and stabilizing after approximately 600 epochs, ultimately achieving an accuracy rate close to 95%. In contrast, the accuracy growth of HACO and ABCO is relatively slow and fluctuates greatly during most training cycles, ultimately stabilizing at around 80% and 75% accuracy. ACO has the lowest accuracy, with limited improvement throughout the entire training process, ultimately stabilizing at around 70%. In Figure 9 (b), the ROC curve of the I-ACO is closest to 1, indicating the optimal classification performance. The I-ACO outperforms other algorithms in both accuracy and classification performance, demonstrating its effectiveness and superiority in optimizing CCL delivery path problems. This indicates that the I-ACO converges faster to higher accuracy and has better generalization ability and robustness in classification tasks.

3.2 Analysis of the optimization effect of CCL delivery path

Table 3 shows the performance of four algorithms in the CCL delivery path

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optimization task. To quantify customer satisfaction indicators, a scoring system based on service level agreements is adopted, which assesses based on the on-time delivery rate of goods and the integrity rate of goods quality. By incorporating the customer satisfaction function into the objective function, the model can optimize the path while minimizing the deviation of customer service levels, thereby enhancing the overall customer satisfaction.

Table 3 Performance of different algorithms in the task of CCL distribution path optimization

Indicators/algorithms	I-ACO	HACO	ABCO	ACO
Accuracy rate (%)	95.2	82.5	73.8	63.4
Run time (seconds)	115	140	175	94
Rate of convergence	0.95	0.83	0.72	0.61
Stability	0.96	0.81	0.75	0.63
CEs (kg)	0.04	0.07	0.09	0.11
Total cost (yuan)	1950	2100	2450	2578
Path length (km)	95	110	130	142
Customer satisfaction (%)	97	89	86	77
Vehicle utilization rate (%)	85	80	75	68

As illustrated in Table 3, the I-ACO algorithm exhibits the highest levels of accuracy, convergence speed, and customer satisfaction, at 95.2%, 0.95%, and 97%, significantly greater than other algorithms. Additionally, the running time of the algorithm is 115 seconds, although not the shortest, considering its high accuracy and

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convergence speed, this running time is acceptable. In terms of CEs, the I-ACO also performs the best, at only 0.04 kg, demonstrating its superiority in environmental protection. In terms of total cost, the I-ACO algorithm leads with a cost of 1,950 yuan, while ACO has the highest cost at 2,578 yuan. In terms of path length, the I-ACO is also the shortest, at 95 km, while ACO is the longest, at 142 km. In terms of vehicle utilization, the I-ACO also has the highest utilization rate, at 85%, indicating that it is more efficient in resource utilization. The I-ACO performs well in multiple key indicators, especially in accuracy and customer satisfaction, demonstrating its effectiveness and reliability in CCL delivery path optimization tasks. The I-ACO provides an efficient, accurate, and environmentally friendly solution for optimizing CCL delivery routes. The error rates of CCL delivery paths under several algorithms are shown in Figure 10.

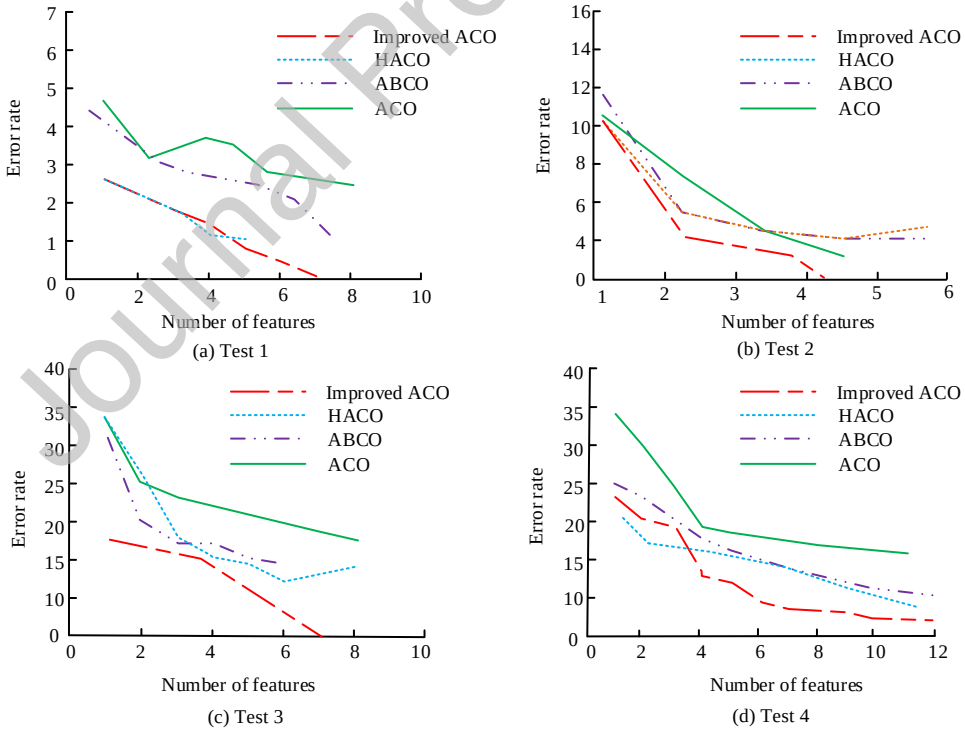

Figure 10 Error rate of CCL distribution path under several algorithms

In Figure 10 (a), on Test Set 1, the I-ACO exhibits the lowest error rate across the entire feature range, especially when the number of features exceeds 4, the error

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rate significantly decreases, and eventually approaches 0. In contrast, HACO and ABCO algorithms have higher error rates, and the improvement with increasing feature count is not as significant as the I-ACO. In Figure 10 (b), on Test Set 2, the I-ACO still maintains the lowest error rate, and the downward trend is the most stable and significant. Figures 10 (c) and (d) show the error rate changes on test sets 3 and 4, with an overall trend similar to the first two test sets. In both test sets, the I-ACO also shows the lowest and continuously decreasing error rate, especially when the amount of features exceeds 6, the error rate almost drops to the lowest. The I-ACO exhibits the optimal CCL delivery path error rate under different feature quantities and test sets, demonstrating its significant advantage in improving delivery path accuracy.

4. Conclusion

This study aimed to address the high costs, high CEs, and low delivery efficiency faced in the CCL delivery process. An I-ACO algorithm was proposed by constructing a CCL delivery path optimization model under LCP to achieve low-carbon, efficient, and economically balanced CCL delivery. In the experiment, the I-ACO performed excellently on multiple key indicators. In terms of accuracy, the improved algorithm achieved 95.2%, significantly higher than other common algorithms. In terms of CEs, the improved algorithm was only 0.04 kg, far lower than other algorithms, demonstrating its superiority in environmental protection. In terms of total cost, the cost of improving the algorithm was 1,950 yuan, which was more advantageous compared to other algorithms. For the path length, the I-ACO had a path length of 95 km, with a customer satisfaction rate of 97% and a vehicle utilization rate of 85%. The I-ACO could effectively reduce logistics costs, reduce CEs, improve delivery efficiency and customer satisfaction, providing an efficient, accurate, and environmentally friendly solution for optimizing CCL delivery routes.

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The CCL distribution route optimization model, by comprehensively considering TWCs, CECs and CDCs, is closely related to the actual CCL distribution scenarios and can effectively address the distribution challenges in reality. The I-ACO algorithm improves the accuracy and adaptability of path selection by introducing DTW technology and chaos theory, and optimizes the quality of the initial population, thereby significantly enhancing the search ability and resolution quality of the algorithm. This is of great significance in practical applications. The algorithm has demonstrated outstanding performance in problem instances of different scales and complexities, which further validates its applicability and effectiveness in actual CCL distribution. The study highlights the significance of adopting data-driven decision-making tools and algorithm optimization in modern logistics management. By leveraging advanced optimization algorithms and models, CCL enterprises can better respond to market changes, enhance operational efficiency, and achieve sustainable development goals. This helps companies enhance their competitiveness and provides technical support and practical guidance for the green transformation and upgrading of the entire industry. Future research can further expand the application scope of the model and take into account more practical constraints, such as traffic congestion and vehicle malfunctions, to enhance the model's practicality and robustness.

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Conflict of Interest

The author confirms that there are no relevant financial or non-financial competing interests to report.

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Conflict of Interest

The author confirms that there are no relevant financial or non-financial competing interests to report.

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